

802.15

A group of wireless personal area network (WPAN) standards from the IEEE for low-power devices in both industrial and consumer applications. See [WPAN](#) and [Internet of Things](#).

802.15.1 - Bluetooth

Working with the Bluetooth special interest group (SIG), the IEEE standardized the lower layers of the Bluetooth specification. Approved by the IEEE in 2002, 802.15.1 is fully compatible with Bluetooth 1.1. See [Bluetooth](#).

802.15.2 - PAN/LAN Co-existence

Explores methods for interoperability between wireless PANs (WPANs) and wireless LANs (WLANs), as well as the capability of transmitting in the same geographic area without interference. See [WPAN](#) and [wireless LAN](#).

802.15.3 - High Data Rate

A WPAN standard for data rates greater than 20 Mbps.

802.15.3a - Very High Data Rate - UWB

The Task Group was dissolved in 2006; however, the IEEE narrowed proposals into two, one of which was multiband OFDM (MB-OFDM), adopted by the WiMedia Alliance for UWB (see [UWB](#)).

802.15.4 - Low Data Rate

Devices in low-rate WPANs (LR-WPANs) run on batteries for years. Nodes can be configured as reduced function devices (RFDs), which communicate with full function devices (FFDs) only. FFDs can communicate with RFDs and FFDs. The 802.15.4 standard specifies only network layers 1 and 2 (see [MAC layer](#)), while examples of the higher-layer protocols are [ZigBee](#), [WirelessHART](#), [Thread](#) and [MiWi](#). See [6LoWPAN](#).

La versión **básica del estándar es 802.15.4a/b**.

- La versión 802.15.4c especifica la capa PHY para China.
- La versión 802.15.4d especifica la capa PHY y MAC para Japón.
- La versión **802.15.4e está orientada a aplicaciones industriales**.
- La versión **802.15.4f especifica la capa PHY y MAC para aplicaciones RFID activas (es decir, con baterías de por medio)**.
- La versión **802.15.4g especifica la capa PHY para redes tipo SUN (Smart Utility Network), es decir Smart Grid**.

802.15.5 - Mesh Networking

Enables mesh networking for 802.15.3 high data rate and 802.15.4 low data rate transmission. See [mesh network](#).

802.15.6 - Body Area Networks

Supports transmission of low-power sensors for health care as well as entertainment. See [BAN](#).

IEEE 802.15.7: Visible Light Communication

As of December 2011, The IEEE 802.15.7 Visible Light Communication **Task Group has completed draft 5c of a PHY and MAC standard for Visible Light Communication (VLC)**. The inaugural meeting for Task Group 7 was held during January 2009, where it was chartered to write standards for free-space optical communication using visible light.

IEEE P802.15.8: Peer Aware Communications

IEEE P802.15.8 received IEEE Standards Board approval on 29 March 2012 to form a Task Group to develop a standard for **Peer Aware Communications (PAC) optimized for peer to peer and infrastructureless communications with fully distributed coordination operating in bands below 11 GHz**. The proposed standard is targeting data rates greater than 100 with scalable data rates up to 10 Mbit/s

IEEE P802.15.9: Key Management Protocol

IEEE P802.15.9 received IEEE Standards Board approval on 7 December 2011 to form a **Task Group to develop a recommended practice for the transport of Key Management Protocol (KMP) datagrams**. The recommended practice will define a message framework based on Information Elements as a transport method for key management protocol (KMP) datagrams and guidelines for the use of some existing KMPs with IEEE Std 802.15.4. The recommended practice will not create a new KMP.[14]

While IEEE Std 802.15.4 has always supported datagram security, it has not provided a mechanism for establishing the keys used by this feature. Lack of key management support in IEEE Std 802.15.4 can result in weak keys, which is a common avenue for attacking the security system. Adding KMP support is critical to a proper security framework. Some of the existing KMPs that it may address are **IETF's PANA**, HIP, IKEv2, IEEE Std 802.1X, and 4-Way-Handshak

IEEE P802.15.10: Layer 2 Routing

IEEE P802.15.10 received IEEE Standards Board approval on 23 August 2013 to form a Task Group to develop of a **recommended practice for routing packets in dynamically changing 802.15.4 wireless networks (changes on the order of a minute time frame)**, with minimal impact to route handling. The goal is to extend the coverage area as the number of nodes increase. **L2R**